CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Arienti, Marco

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0001-8166-0016

*POSITION TITLE: Principal Member of Technical Staff

*ORGANIZATION AND LOCATION: Sandia National Laboratories, Livermore, California, United States

Proposals and Active Projects

***Proposal/Active Project Title:** Multiphysics Modeling and Simulation for Energy Innovation

***Status of Support:** Current

Proposal/Award Number:

***Source of Support:** DOE 241660

***Primary Place of Performance:** SNL

***Proposal/Active Project Start Date: (MM/YYYY):** 01/2026

***Proposal/Active Project End Date: (MM/YYYY):** 01/2027

***Total Anticipated Proposal/Project Amount:** $2,970,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	6
2027	1

***Overall Objectives:**

Perform high fidelity modeling and simulation for energy, aerospace, and defense applications using technology and data derived from DOE WETO programs.

***Statement of Potential Overlap:**

Project work development is synergistic and does not conflict with proposed GENESIS work.

***Proposal/Active Project Title:** High-fidelity modeling of particle streamers from high explosive testing

***Status of Support:** Current

Proposal/Award Number:

***Source of Support:** NNSA

***Primary Place of Performance:** SNL

***Proposal/Active Project Start Date: (MM/YYYY):** 11/2024

***Proposal/Active Project End Date: (MM/YYYY):** 09/2026

***Total Anticipated Proposal/Project Amount:** $1,350,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.1

***Overall Objectives:**

In response to the Proliferation Detection R&D NA-22 call of 2024, we have proposed to develop a physics-driven, end-to-end modeling framework for the metal streamers formed in certain detonation fireballs. This capability is necessary to support our ability to characterize and discriminate between conventional explosions and weapons development activities. A basic understanding of metal particulate optical emission requires the knowledge of the surface state, temperature, and composition, because distinct combustion regimes exist depending on these parameters as well as the particulate size distribution. The project generates submodels for the description of metal particulates that can be coupled to existing software platforms for event-scale simulations at mission-relevant conditions.

***Statement of Potential Overlap:**

No overlap.

***Proposal/Active Project Title:** AI-Agent Orchestrated CFD and Generative Surrogates for Rapid Fluid Flow Design and Control

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** DE-FOA-0003612

***Primary Place of Performance:** SNL

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2026

***Proposal/Active Project End Date: (MM/YYYY):** 03/2027

***Total Anticipated Proposal/Project Amount:** $750,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1
2026	1

***Overall Objectives:**

Accelerate the design of critical energy components and applications, and provide a fast-running accurate model for control with a combination of AI techniques that target workflow automation and generative AI for fluid flow prediction with complex geometries.

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:** Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** DE-FOA-0003612

***Primary Place of Performance:** SNL

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2026

***Proposal/Active Project End Date: (MM/YYYY):** 03/2027

***Total Anticipated Proposal/Project Amount:** $495,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1
2027	1

***Overall Objectives:**

The objective will be to develop AI-enhanced sharp-interface numerical methods that can reduce the computational cost in R&D of advanced combustion devices (such as jet engines and rotating detonation rockets). This activity will take place using existing DOE-developed simulation codes.

***Statement of Potential Overlap:**

None

Addendum

Do the Notice of Funding Opportunity (NOFO) or Award instructions require a Current and Pending (Other) Support addendum?

No

Certification:

I understand that I have been designated as a covered individual by the Federal funding agency.

I certify to the best of my knowledge and belief that the information contained in this Current and Pending (Other) Support Form is true, complete, and accurate. I understand that any false, fictitious, or fraudulent information, misrepresentations, half-truths, or omissions of any material fact, may subject me to criminal, civil or administrative penalties for fraud, false statements, false claims or otherwise. (18 U.S.C. §§ 1001 and 287, and 31 U.S.C. 3729-3733 and 3801-3812). I further understand and agree that (1) the statements and representations made herein are material to DOE's funding decision, and (2) I have a responsibility to update the disclosures during the project period of the award should circumstances change which impact the responses provided above.

I also certify that, at the time of submission, I am not a party in a malign foreign talent recruitment program. I further understand should I take action to involve myself with a Malign Foreign Talent Recruitment Program during the period of performance of the award, I must notify the recipient's Authorized Agent immediately, but no later than five business days of taking such action and immediately recuse myself from all DOE awards.

If this is an R&D proposal/award, I also certify to the following statement:

Within the past 12 months I have completed research security training meeting the requirements in SEC. 10634(b) of 42 USC 19234.

Certified by Arienti, Marco in SciENcv on 2026-04-14 01:26:17